

LAMIN JAWNEH

Trainee Data Scientist · ML Researcher · MSc Computer Science

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RESEARCH PROFILE

Early-career data scientist and ML researcher with an MSc in Computer Science from Constructor University (Bremen, Germany) and 5+ years of applied experience across machine learning, software engineering, and systems infrastructure. Currently conducting research at the Medical Research Council Unit The Gambia (MRCG), one of Africa's leading biomedical research institutions, developing a transfer learning framework for clinical outcome prediction in low-resource African health settings — with a target publication in npj Digital Medicine or The Lancet Digital Health. Combines deep technical ML expertise with rare applied experience in healthcare AI, financial systems, and software development. Seeking a doctoral position to advance research at the intersection of machine learning and real-world human-centred systems.

RESEARCH EXPERIENCE

Trainee Data Scientist & ML Researcher | **Medical Research Council Unit The Gambia (MRCG)** — Fajara, The Gambia *March 2026 – Present*

- Designing and implementing a transfer learning framework that pre-trains deep learning models on MIMIC-III (46,000+ ICU patients) and fine-tunes on small African clinical cohorts, directly addressing the data scarcity problem in LMIC health AI.
- Evaluating fine-tuning strategies (full fine-tuning, layer freezing, adapter modules) against baselines including logistic regression, XGBoost, and from-scratch deep learning.
- Conducting domain shift analysis between MIMIC-III and African cohort data using distributional comparison methods.
- **npj Digital Medicine** or The Lancet Digital Health, targeted for submission within 5–6 months.
- Working with transformer-based tabular architectures (FT-Transformer, TabTransformer), PyTorch, and statistical validation frameworks (AUROC, AUPRC, calibration, bootstrap CIs).

PROFESSIONAL EXPERIENCE

IT Manager | **AfricMed International Hospital** — Brusubi, The Gambia *Jan 2026 – Present*

- Manage hospital IT infrastructure (servers, network, cloud, workstations) ensuring high availability for HIS/EMR, laboratory, pharmacy, and billing systems.
- Implement and enforce IT security policies: role-based access control, data encryption, endpoint protection, and regular system audits to safeguard patient data.
- Oversee backup and disaster recovery processes to ensure business continuity.

Software Developer | **Security Technology Alarm** — Karaba Avenue, The Gambia *Sep 2025 – Present*

- Design and develop scalable backend systems and RESTful APIs using Java, Python, and modern frameworks.
- Build responsive web applications and manage deployment across development, staging, and production environments.
- Database schema design, data integration, and query optimisation (PostgreSQL, MySQL).

System Administrator | Trust Bank — Banjul, The Gambia *Apr 2022 – Sep 2025*

- Administered core banking IT infrastructure including servers, networks, and workstations across the organisation.
- Provided advanced technical support for SWIFT, RTGS, ACP/ACH, Online Banking, and PAPSS payment systems, ensuring secure interbank transactions.
- Managed Microsoft Active Directory, Exchange Server, and Group Policy; configured and maintained LAN/WAN, routers, firewalls, and switches.
- Supported disaster recovery and backup operations for critical financial systems.

Data Science Practitioner | Cangam Analytics — Sukuta, The Gambia *Jan 2020 – Mar 2022*

- Developed predictive models using TensorFlow, Keras, and Scikit-learn to support strategic business decisions.
- Built customer analytics models: churn prediction, lifetime value estimation, and customer segmentation.
- Presented insights and dashboards to management using Power BI and Matplotlib.

IT Lecturer | Suna Institute of Science and Technology — Brikama, The Gambia *Jul 2021 – Mar 2022*

- Delivered lectures on computer hardware, software systems, and network fundamentals to diploma and certificate students.
- Designed and supervised hands-on lab sessions covering operating systems and IT support.

EDUCATION

MSc — Computer Science & Software Engineering | Constructor University — Bremen, Germany *Sep 2023 – Jun 2025*

- Thesis: Enhancing Transaction Reliability and Security in Real-time Financial Systems Through Dynamic Software Update Methodologies

BSc — Computer Science | University of The Gambia — Kanifing, The Gambia *2017 – 2020*

Diploma — Information Technology | Lasting Solutions — Kanifing, The Gambia *Jan 2016 – Nov 2016*

TECHNICAL SKILLS

Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn — deep learning, transfer learning, tabular transformers, survival analysis
Data Science	Statistical modelling, predictive analytics, customer segmentation, Power BI, Matplotlib, Pandas, NumPy
Data Engineering	SQL, PostgreSQL, MySQL, ETL pipelines, database schema design, data integration
Languages	Python, Java, SQL, JavaScript — production and research contexts
Software Dev	RESTful APIs, React, Spring Boot, Git/GitHub, CI/CD, Jira, Trello, Asana
Systems & Infra	Windows Server, Linux, VMware, Hyper-V, LAN/WAN, TCP/IP, DNS, VPN, VLANs, Cisco/Juniper
Security	RBAC, data encryption, endpoint protection, SWIFT/RTGS/PAPSS payment security, system audits
Research Tools	MIMIC-III (PhysioNet), FT-Transformer, TabTransformer, domain adaptation, bootstrap validation

CURRENT RESEARCH PROJECT

Transfer Learning for Clinical Prediction in Low-Data African Health Settings

MRCG Data Science Department | Target: npj Digital Medicine / The Lancet Digital Health | Timeline: 5–6 months

- Proposes a novel transfer learning framework pre-trained on MIMIC-III and fine-tuned on small African clinical cohorts to address the data scarcity gap in LMIC health AI.
- Benchmarks transfer learning strategies against classical ML baselines under conditions typical of sub-Saharan African health data.
- Quantifies domain shift between Western ICU data and African clinical settings — a gap not previously studied in the literature.
- All code will be open-sourced for use by other African health institutions.

LANGUAGES

English	Professional fluency — primary language of research and publication
Mandinka / Wolof	Native

REFERENCES

Available on request — including supervisor at MRCG Data Science Department.